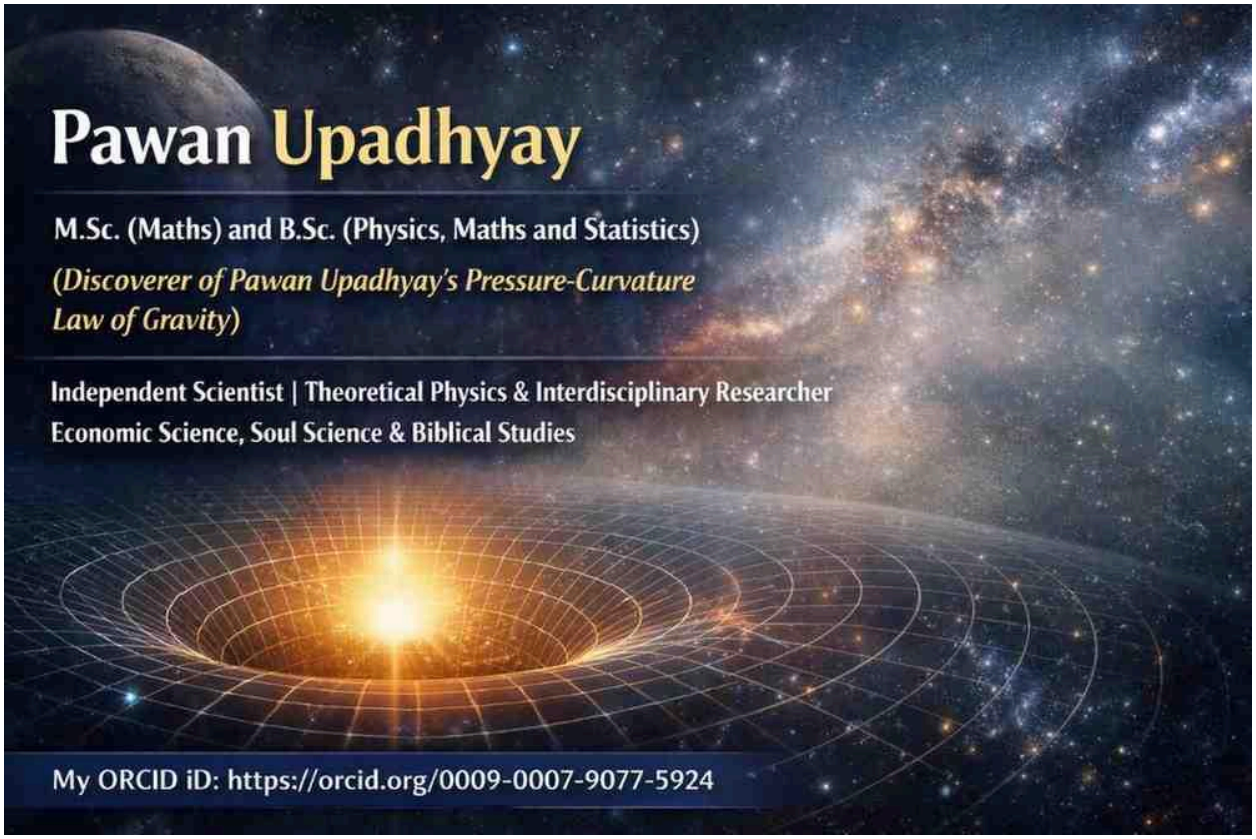
A SPACETIME FIELD FORMULATION OF ENERGY DENSITY, GRAVITATIONAL PRESSURE, AND PRESSURE-GRADIENT FORCE DENSITY

Author: Pawan Upadhyay
Affiliation: Independent Researcher and Independent Scientist
Email:pawanupadhyay28@hotmail.com
2nd Email: pawankanyakubj@gmail.com
ORCID iD: https://orcid.org/0009-0007-9077-5924



ABSTRACT

This paper proposes a spacetime-field formulation in which mass density, energy density, equation-of-state parameter, gravitational pressure, pressure gradient, and gravitational pressure-force density are treated as local fields rather than as globally constant quantities.

The spacetime coordinate is defined as:

$x^\mu = (x, y, z, ct)$

where $\mu = 0, 1, 2, 3$.

The proposed energy-density field is related to the mass-density field through:

$E_d(x^\mu) = \rho(x^\mu)c^2$

A spacetime-dependent equation-of-state parameter field, $\omega(x^\mu)$, is introduced to couple the energy-density field to a proposed gravitational-pressure field according to:

$P_g(x^\mu) = \omega(x^\mu)E_d(x^\mu)$

Consequently:

$P_g(x^\mu) = \omega(x^\mu)\rho(x^\mu)c^2$

The spatial gradient of this pressure field is defined as:

$\nabla P_g(x^\mu)$

and the associated proposed gravitational pressure-force-density field is:

$F(x^\mu) = -\nabla P_g(x^\mu)$

Applying the product rule gives:

$F(x^\mu) = -c^2[\rho(x^\mu)\nabla\omega(x^\mu) + \omega(x^\mu)\nabla\rho(x^\mu)]$

The formulation therefore allows the proposed gravitational pressure-force density to depend on both spatial variations of mass density and spatial variations of the equation-of-state parameter.

The paper defines the mathematical structure of these fields and discusses their interpretation, dimensional consistency, limiting cases, and relationship to established relativistic field theories. The proposed framework is presented as a theoretical model requiring further mathematical development and empirical investigation.

KEYWORDS

Spacetime field; mass density; energy density; gravitational pressure; equation of state; pressure gradient; force density; spacetime-dependent coupling.

1. INTRODUCTION

Physical quantities describing matter and energy can be represented by fields. A field assigns a physical quantity to every point in space and, in relativistic physics, to every event in spacetime.

In this paper, the fundamental spacetime coordinate is written as:

$x^\mu = (x, y, z, ct)$

where x, y, and z are spatial coordinates, t is time, c is the speed of light, and $\mu = 0, 1, 2, 3$.

The central proposition of this work is that quantities associated with mass, energy, pressure, and their coupling may be represented as local spacetime-dependent fields.

The mass density is therefore written as:

$\rho = \rho(x^\mu)$

The energy density is:

$E_d = E_d(x^\mu)$

The equation-of-state parameter is:

$\omega = \omega(x^\mu)$

and the proposed gravitational pressure is:

$P_g = P_g(x^\mu)$

The central relation proposed in this framework is:

$P_g(x^\mu) = \omega(x^\mu)E_d(x^\mu)$

When the energy-density field is related to the mass-density field by:

$E_d(x^\mu) = \rho(x^\mu)c^2$

the gravitational-pressure field becomes:

$P_g(x^\mu) = \omega(x^\mu)\rho(x^\mu)c^2$

The spatial variation of this pressure field provides the basis for defining a proposed gravitational pressure-force-density field.

2. SPACETIME COORDINATE

The spacetime coordinate used throughout this formulation is:

$x^\mu = (x, y, z, ct)$

The four components are:

$x^0 = ct$

$x^1 = x$

$x^2 = y$

$x^3 = z$

Every physical quantity introduced in the model is considered a function of this spacetime coordinate.

Therefore:

$\rho = \rho(x^\mu)$

$E_d = E_d(x^\mu)$

$\omega = \omega(x^\mu)$

$P_g = P_g(x^\mu)$

The notation emphasizes that the value of a field may differ from one spacetime point to another.

For four different spacetime points A, B, C, and D:

$\rho_A = \rho(x_A^\mu)$

$\rho_B = \rho(x_B^\mu)$

$\rho_C = \rho(x_C^\mu)$

$\rho_D = \rho(x_D^\mu)$

In general:

$\rho_A \neq \rho_B \neq \rho_C \neq \rho_D$

The same local-field principle applies to energy density, gravitational pressure, equation-of-state parameter, pressure gradient, and force density.

3. MASS DENSITY FIELD

The mass density field is defined as the scalar spacetime field:

$\rho(x^\mu)$

It represents the local mass per unit spatial volume at a given spacetime event.

The field formulation permits mass density to vary spatially and temporally:

$\rho = \rho(x, y, z, ct)$

Therefore, different spatial locations may possess different local mass densities at the same coordinate time.

The spatial variation of the mass-density field is represented by:

$\nabla \rho(x^\mu)$

The fundamental concept is that mass density is treated as a local field rather than as a single universal number.

4. ENERGY DENSITY FIELD

The energy density field is defined as:

$E_d(x^\mu)$

It represents the local energy per unit spatial volume associated with the field configuration.

Within the proposed formulation, the local mass-energy relation is:

$E_d(x^\mu) = \rho(x^\mu)c^2$

This means that the mass-energy relation is applied locally to the mass-density field.

At point A:

$E_{d,A} = \rho_A c^2$

At point B:

$E_{d,B} = \rho_B c^2$

At point C:

$E_{d,C} = \rho_C c^2$

At point D:

$E_{d,D} = \rho_D c^2$

Therefore, variation in the mass-density field produces corresponding variation in the energy-density field.

5. EQUATION-OF-STATE PARAMETER FIELD

The equation-of-state parameter is introduced as the dimensionless scalar spacetime field:

$\omega(x^\mu)$

It specifies the local proportionality between the proposed gravitational pressure and energy density.

The fundamental relation is:

$P_g(x^\mu) = \omega(x^\mu)E_d(x^\mu)$

When $E_d(x^\mu)$ is non-zero, the equation-of-state parameter can be written as:

$\omega(x^\mu) = P_g(x^\mu) / E_d(x^\mu)$

Unlike a constant equation-of-state parameter, the present formulation permits:

$\omega = \omega(x, y, z, ct)$

Therefore:

ω_A

ω_B

ω_C

ω_D

may all have different values at different spacetime points.

The field $\omega(x^\mu)$ consequently represents a local equation-of-state parameter rather than a universal constant.

6. GRAVITATIONAL PRESSURE FIELD

The proposed gravitational pressure field is defined as the scalar spacetime field:

$P_g(x^\mu)$

Its fundamental definition is:

$P_g(x^\mu) = \omega(x^\mu)E_d(x^\mu)$

Substituting:

$E_d(x^\mu) = \rho(x^\mu)c^2$

gives:

$P_g(x^\mu) = \omega(x^\mu)\rho(x^\mu)c^2$

This is the central constitutive relation of the proposed framework.

At a particular spacetime point A:

$P_{g,A} = \omega_A \rho_A c^2$

At another spacetime point B:

$P_{g,B} = \omega_B \rho_B c^2$

Thus, gravitational pressure may differ between points because the mass density, equation-of-state parameter, or both may differ.

7. ENERGY-DENSITY–GRAVITATIONAL-PRESSURE COUPLING FIELD

The proposed coupling between energy density and gravitational pressure is represented by:

$P_g(x^\mu) = \omega(x^\mu) E_d(x^\mu)$

In this formulation, $\omega(x^\mu)$ acts as a dimensionless local coupling parameter.

The coupling field may therefore be represented as:

$C_{EP}(x^\mu) = \omega(x^\mu)$

The gravitational pressure becomes:

$P_g(x^\mu) = C_{EP}(x^\mu) E_d(x^\mu)$

Using the mass-energy density relation:

$P_g(x^\mu) = C_{EP}(x^\mu) \rho(x^\mu) c^2$

With the identification:

$C_{EP}(x^\mu) = \omega(x^\mu)$

the coupling parameter and equation-of-state parameter are mathematically equivalent within this formulation.

8. PRESSURE GRADIENT FIELD

The pressure gradient field is defined as the spatial gradient of the gravitational pressure field:

$\nabla P_g(x^\mu)$

Explicitly:

$\nabla P_g(x^\mu) = (\partial P_g/\partial x, \partial P_g/\partial y, \partial P_g/\partial z)$

The pressure gradient is a vector field. It specifies the direction and rate of maximum spatial increase of the gravitational-pressure field.

Starting from:

$P_g = \omega E_d$

application of the product rule gives:

$\nabla P_g = E_d \nabla \omega + \omega \nabla E_d$

Since:

$E_d = \rho c^2$

and c is constant:

$\nabla E_d = c^2 \nabla \rho$

Therefore:

$\nabla P_g = c^2[\rho \nabla \omega + \omega \nabla \rho]$

This equation demonstrates that the pressure gradient has two contributions:

1. Equation-of-state contribution: $c^2 \rho \nabla \omega$
2. Mass-density contribution: $c^2 \omega \nabla \rho$

Therefore:

$\nabla P_g = c^2 \rho \nabla \omega + c^2 \omega \nabla \rho$

9. GRAVITATIONAL PRESSURE-FORCE-DENSITY FIELD

The proposed gravitational pressure-force-density field is defined by:

$F(x^\mu) = -\nabla P_g(x^\mu)$

Substituting the pressure-gradient relation gives:

$F(x^\mu) = -c^2[\rho(x^\mu) \nabla \omega(x^\mu) + \omega(x^\mu) \nabla \rho(x^\mu)]$

The negative sign indicates that the proposed force-density field is directed toward decreasing gravitational pressure.

The quantity F_g in this formulation is a force per unit volume associated with the proposed gravitational-pressure field.

It should therefore be distinguished from the total gravitational force acting on a finite object.

10. LIMITING CASES

10.1 CONSTANT EQUATION-OF-STATE PARAMETER

If:

$$\nabla \omega = 0$$

then:

$$F = -\omega c^2 \nabla \rho$$

In this case, the proposed pressure-force-density field is generated entirely by spatial variation of mass density.

10.2 UNIFORM MASS DENSITY

If:

$$\nabla \rho = 0$$

then:

$$F = -\rho c^2 \nabla \omega$$

In this case, the proposed pressure-force-density field is generated entirely by spatial variation of the equation-of-state parameter.

10.3 UNIFORM MASS DENSITY AND CONSTANT EQUATION OF STATE

If:

$$\nabla \rho = 0$$

and:

$$\nabla \omega = 0$$

then:

$$\nabla P_g = 0$$

and therefore:

$F = 0$

Thus, within the proposed pressure-gradient mechanism, no pressure-gradient force density arises from a spatially uniform gravitational-pressure field.

11. FIELD VALUES AT DIFFERENT SPACETIME POINTS

Consider four spacetime points A, B, C, and D.

The local fields are:

$\rho_A, \rho_B, \rho_C, \rho_D$

$E_{d,A}, E_{d,B}, E_{d,C}, E_{d,D}$

$\omega_A, \omega_B, \omega_C, \omega_D$

and:

$P_{g,A}, P_{g,B}, P_{g,C}, P_{g,D}$

At each point:

$E_{d,i} = \rho_i c^2$

and:

$P_{g,i} = \omega_i E_{d,i}$

where i represents A, B, C, or D.

Therefore:

$P_{g,i} = \omega_i \rho_i c^2$

The field values are consequently local properties of spacetime rather than single global constants.

12. DIMENSIONAL ANALYSIS

Mass density has dimensions of:

kg/m^3

The speed of light squared has dimensions of:

m^2/s^2

Therefore:

$\rho c^2 = kg/(m \cdot s^2)$

which is equivalent to:

J/m^3

Energy density and pressure possess the same physical dimensions:

$J/m^3 = N/m^2$

Therefore:

$[E_d] = [P_g]$

The equation-of-state parameter ω is dimensionless:

$[\omega] = 1$

The pressure gradient has dimensions of:

N/m^3

which corresponds to force per unit volume.

Therefore, the proposed force-density field has dimensions:

$[F] = N/m^3$

The basic dimensional structure of the proposed equations is therefore internally consistent.

13. MATHEMATICAL STRUCTURE OF THE MODEL

The proposed model can be summarized as the following field sequence:

$\rho(x^\mu)$

$\rightarrow E_d(x^\mu)$

$\rightarrow P_g(x^\mu)$

$\rightarrow \nabla P_g(x^\mu)$

$\rightarrow F(x^\mu)$

More explicitly:

$\rho(x^\mu) \rightarrow E_d(x^\mu) = \rho(x^\mu)c^2$

$E_d(x^\mu) \rightarrow P_g(x^\mu) = \omega(x^\mu)E_d(x^\mu)$

$P_g(x^\mu) \rightarrow \nabla P_g(x^\mu)$

$\nabla P_g(x^\mu) \rightarrow F_g(x^\mu) = -\nabla P_g(x^\mu)$

The complete proposed relation is:

$F(x^\mu) = -\nabla[\omega(x^\mu)\rho(x^\mu)c^2]$

and therefore:

$F(x^\mu) = -c^2[\rho(x^\mu)\nabla\omega(x^\mu) + \omega(x^\mu)\nabla\rho(x^\mu)]$

14. RELATION TO RELATIVISTIC FIELD THEORY

The use of:

$x^\mu = (x, y, z, ct)$

places the formulation in spacetime-field notation.

However, the use of spacetime-dependent scalar fields alone does not constitute a complete relativistic theory of gravitation.

In general relativity, gravitation is described geometrically through spacetime curvature, with matter and energy represented by the stress-energy tensor.

Pressure can contribute to the stress-energy tensor, and relativistic fluids may be described using energy density, pressure, four-velocity, and an equation of state.

The present formulation proposes a particular scalar pressure relation:

$P_g = \omega E_d$

together with the pressure-gradient relation:

$F = -\nabla P_g$

Therefore, additional theoretical work is required to establish how these proposed fields relate to the metric tensor, stress-energy tensor, conservation laws, and relativistic covariance.

So Read full theory of Proposed Pawan Upadhyay's Pressure Curvature Law of Gravity here

https://archive.org/details/pawan-upadhyay-s-pressure-curvature-law-of-gravity

15. DISCUSSION

The principal feature of the proposed framework is the treatment of physical quantities as local spacetime fields.

The mass density is:

$\rho = \rho(x^\mu)$

The energy density is:

$E_d = \rho(x^\mu)c^2$

The equation-of-state parameter is:

$\omega = \omega(x^\mu)$

The gravitational pressure is:

$P_g = \omega(x^\mu)\rho(x^\mu)c^2$

The pressure gradient is:

$\nabla P_g = c^2[\rho \nabla \omega + \omega \nabla \rho]$

This decomposition separates the pressure gradient into two contributions:

1. A mass-density-gradient contribution:

$c^2\omega \nabla \rho$

2. An equation-of-state-gradient contribution:

$c^2\rho \nabla \omega$

This is an important property of the model.

A pressure gradient can exist because of a spatial variation in mass density:

$\nabla \rho \neq 0$

or because of a spatial variation in the equation-of-state parameter:

$\nabla \omega \neq 0$

or because both quantities vary simultaneously.

Therefore, the proposed pressure-gradient force-density field can be expressed as:

$$F_g = -c^2\omega\nabla\rho - c^2\rho\nabla\omega$$

This provides a mathematical framework for investigating whether local variations in mass-energy density and equation-of-state behavior can be associated with a gravitational-pressure field.

16. LIMITATIONS AND FUTURE DEVELOPMENT

The present formulation is a theoretical proposal and requires further investigation before any claim of a new gravitational law can be established.

First, the physical meaning of the proposed gravitational pressure P_g must be precisely defined and distinguished from ordinary thermodynamic pressure.

Second, the relation:

$$F = -\nabla P_g$$

must be derived from a fundamental dynamical principle if the model is intended to represent an actual gravitational interaction.

Third, the transformation properties of the proposed fields under Lorentz transformations and general coordinate transformations must be established.

Fourth, the model must be connected consistently with the relativistic stress-energy tensor.

Fifth, conservation laws must be investigated.

Sixth, the model should be compared with established gravitational theories, particularly Newtonian gravity and general relativity.

Finally, the theory must produce testable predictions that can be compared with astronomical, cosmological, or laboratory observations.

17. CONCLUSION

This paper has presented a spacetime-field formulation based on:

$$x^\mu = (x, y, z, ct)$$

The formulation treats mass density, energy density, equation-of-state parameter, gravitational pressure, pressure gradient, and gravitational pressure-force density as local spacetime-dependent fields.

The fundamental energy-density relation is:

$$E_d(x^\mu) = \rho(x^\mu)c^2$$

The proposed gravitational-pressure relation is:

$P_g(x^\mu) = \omega(x^\mu)E_d(x^\mu)$

Therefore:

$P_g(x^\mu) = \omega(x^\mu)\rho(x^\mu)c^2$

The pressure-gradient field is:

$\nabla P_g(x^\mu) = c^2[\rho(x^\mu)\nabla\omega(x^\mu) + \omega(x^\mu)\nabla\rho(x^\mu)]$

The proposed gravitational pressure-force-density field is:

$F(x^\mu) = -\nabla P_g(x^\mu)$

Therefore:

$F(x^\mu) = -c^2[\rho(x^\mu)\nabla\omega(x^\mu) + \omega(x^\mu)\nabla\rho(x^\mu)]$

The central result is that the proposed pressure-force-density field contains two mathematical sources: variation in mass density and variation in the equation-of-state parameter.

The framework therefore provides a starting point for further investigation into a possible spacetime-dependent relationship among mass density, energy density, gravitational pressure, equation-of-state behavior, pressure gradients, and force density.

A complete physical theory would require a covariant formulation, dynamical field equations, conservation laws, compatibility with established gravitational physics, and experimentally or observationally testable predictions.

REFERENCES

1. Einstein, A. (1905). On the Electrodynamics of Moving Bodies.
2. Einstein, A. (1915). The Field Equations of Gravitation.
3. Misner, C. W., Thorne, K. S., and Wheeler, J. A. (1973). Gravitation. W. H. Freeman.
4. Weinberg, S. (1972). Gravitation and Cosmology. John Wiley & Sons.
5. Landau, L. D., and Lifshitz, E. M. The Classical Theory of Fields. Butterworth-Heinemann.
6. Carroll, S. M. (2004). Spacetime and Geometry: An Introduction to General Relativity. Addison-Wesley.

7. Upadhyay,P.(2025). Pawan Upadhyay's Pressure Curvature Law of Gravity (PPC Law of Gravity) and Gravitational Pressure waves https://archive.org/details/@pawan_upadhyay

CORE PROPOSED EQUATIONS

$x^{\mu} = (x, y, z, ct)$

$E_d(x^{\mu}) = \rho(x^{\mu})c^2$

$P_g(x^{\mu}) = \omega(x^{\mu})E_d(x^{\mu})$

$P_g(x^{\mu}) = \omega(x^{\mu})\rho(x^{\mu})c^2$

$\nabla P_g(x^{\mu}) = c^2[\rho(x^{\mu})\nabla\omega(x^{\mu}) + \omega(x^{\mu})\nabla\rho(x^{\mu})]$

$F(x^{\mu}) = -\nabla P_g(x^{\mu})$

$F(x^{\mu}) = -c^2[\rho(x^{\mu})\nabla\omega(x^{\mu}) + \omega(x^{\mu})\nabla\rho(x^{\mu})]$